

SLAVGOROD, Russian Federation

WMO: 299150

Lat: 52.967N

Lon: 78.650E

Elev: 413

StdP: 14.48

Time Zone: 7.00 (E07)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.5	-25.6	-36.8	0.7	-31.4	-30.6	1.0	-25.5	24.6	24.8	21.0	19.8	4.3	340	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	89.9	64.6	86.7	64.4	83.5	63.4	68.6	81.5	67.1	79.8	65.7	78.4	8.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.6	92.6	72.7	62.7	86.6	71.7	61.1	81.7	70.5	33.2	81.7	31.9	79.4	30.8	78.5	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.2	17.6	15.6	-34.3	96.1	7.3	4.0	-39.5	98.9	-43.8	101.3	-47.9	103.5	-53.2	106.4		
			-34.5	71.4	7.2	3.0	-39.7	73.5	-43.9	75.2	-47.9	76.9	-53.2	79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	-0.4	4.3	19.4	42.5	56.3	67.2	69.5	65.6	54.0	39.9	21.9	7.7
	DBStd	27.04	14.82	12.75	12.27	10.58	9.66	7.64	6.27	7.03	8.71	9.28	13.85	13.92
	HDD50	6661	1563	1281	948	270	50	2	0	1	55	335	844	1312
	HDD65	10484	2028	1701	1413	676	300	65	27	79	345	779	1294	1777
	CDD50	2093	0	0	0	45	244	518	606	486	174	20	0	0
	CDD65	442	0	0	0	1	29	132	168	98	14	0	0	0
	CDH74	4603	0	0	0	31	417	1355	1536	1051	211	2	0	0
	CDH80	1545	0	0	0	5	120	475	530	346	69	0	0	0
	WSAvg	7.7	7.6	7.8	8.3	8.6	8.3	7.3	6.5	6.7	7.1	7.5	8.2	8.6
	PrecAvg	11.7	0.6	0.5	0.6	0.6	0.9	1.5	2.2	1.3	0.9	1.0	0.9	0.7
Precipitation	PrecMax	17.0	1.3	1.2	1.7	1.5	3.1	5.0	6.0	3.5	1.8	2.6	2.2	1.3
	PrecMin	6.9	0.1	0.0	0.1	0.0	0.1	0.4	0.2	0.1	0.2	0.2	0.4	0.2
	PrecStd	2.8	0.3	0.3	0.4	0.4	0.6	1.2	1.3	0.7	0.4	0.7	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	33.8	49.5	78.3	89.8	93.0	93.5	94.2	88.3	70.0	50.4	35.1
		MCWB	33.0	32.3	42.1	57.1	62.1	65.1	67.3	65.9	62.5	53.0	44.1	33.1
	2%	DB	29.7	30.6	41.9	71.0	83.1	89.0	89.1	88.1	79.8	63.7	45.4	32.5
		MCWB	28.3	29.4	37.7	54.4	59.5	63.5	66.5	64.7	58.9	50.5	41.0	30.9
	5%	DB	25.7	27.1	37.3	65.5	78.8	85.4	85.8	83.5	74.2	58.2	41.1	29.7
		MCWB	24.5	25.9	34.6	50.6	58.1	63.3	65.8	63.8	56.4	48.0	37.8	28.4
	10%	DB	21.8	23.4	34.7	60.1	74.1	81.9	82.4	79.9	68.9	53.2	37.3	26.4
		MCWB	20.7	22.4	32.7	48.1	56.1	62.4	65.0	62.2	54.3	45.5	35.1	25.2
	0.4%	WB	33.4	32.4	43.0	58.4	64.3	68.7	71.1	69.7	64.0	54.7	45.8	33.6
		MCDB	34.1	33.3	48.6	73.9	83.2	84.2	83.9	81.9	84.0	65.6	48.9	34.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.6	29.3	38.1	54.9	61.6	66.7	69.3	67.0	60.4	51.6	41.1	31.2
		MCDB	29.7	30.4	41.8	69.0	78.5	80.8	82.0	81.6	74.5	61.4	44.9	32.4
	5%	WB	24.6	26.0	35.0	51.8	59.5	65.3	68.0	65.4	57.9	48.9	38.1	28.4
		MCDB	25.7	27.1	37.0	63.4	74.9	78.9	79.8	79.8	71.4	56.8	40.9	29.6
	10%	WB	20.7	22.3	32.9	48.9	57.3	64.0	66.6	63.8	55.5	46.1	35.1	25.1
		MCDB	21.8	23.4	34.6	59.0	71.4	77.3	78.0	77.0	66.8	52.7	37.3	26.4
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.7	15.5	19.9	22.1	20.2	18.9	20.6	20.0	15.8	12.3	14.4
		MCDBR	16.9	14.8	14.7	28.3	28.4	25.6	23.3	26.4	28.2	24.5	14.0	16.4
		MCWBR	16.6	14.5	11.6	14.4	11.8	8.7	7.1	9.0	11.6	13.1	10.4	16.1
	5% WB	MCDBR	17.1	15.1	14.0	26.1	24.4	20.9	19.0	22.5	25.1	22.2	13.2	16.3
		MCWBR	16.7	14.7	11.4	14.4	11.8	8.4	7.2	8.8	11.4	13.1	10.3	16.1
Clear-Sky Solar Irradiance	taub		0.298	0.331	0.386	0.407	0.410	0.414	0.440	0.420	0.375	0.351	0.334	0.305
	taud		2.026	1.963	1.927	2.153	2.196	2.271	2.201	2.223	2.296	2.260	2.113	1.995
	Ebn at Noon		202	232	245	260	266	266	256	253	252	229	187	170
	Edh at Noon		25	36	47	43	43	40	43	39	32	27	23	21
All-Sky Solar Radiation	RadAvg		367	720	1187	1507	1879	2003	1810	1571	1131	608	301	246
	RadStd		32	34	95	107	146	134	121	109	95	78	34	36

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	-0.83	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page